

RMG-1: Relational Medium Gravity

Phantom-free relational gravity from a stable base-medium

Internal core draft, version 6.2

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Purpose

RMG is the phantom-free rebuild of the previous gravitational line.

Core sentence:

gravity is the low-energy relational geometry of a stable base-medium.

RMG does not start from a fundamental phantom scalar field. It starts from a base-medium $\mathcal{M}$ and treats the tested GR/ADM structure as the low-energy fixed point that must emerge.

The protected low-energy phase admits only operational relational observables, not a separate local medium-frame field.

A central core gate is that all physical low-energy principal operators share one kinetic metric kernel.

Status labels

Every statement in RMG-1 must have a status:

Primitive Postulate

minimal starting property of the base-medium;

Primitive Protection Principle

low-energy relational rule that forbids dangerous operators;

Working Assumption

temporary bridge used inside RMG-1, still requiring deeper derivation;

Result

statement derived from the listed assumptions;

Low-Energy Target

structure required by observation and used as the safe fixed point;

Open Theorem

claim still to be proven from $\mathcal{M}$;

Failure Gate

condition whose failure damages the core theory.

1 Definition

RMG = Relational Medium Gravity.

Working definition:

RMG is a gravitational framework in which a single base-medium $\mathcal{M}$ gives rise to matter modes, rods, clocks, signals and operational geometry. The deeper task is to explain why the universal low-energy geometry is the ADM/GR fixed point.

2 Core Terms

$\mathcal{M}$

is the base-medium. It is not assumed to be an ordinary scalar field.

$\mathcal{W}$

is the relational medium-history.

X_F

is a foliation or embedding description of the same history $\mathcal{W}$.

$s(x)$

is the local operational factor shared by clocks, rods and signals in the weak-sector bridge description.

u

is the compression/lapse potential:

$$s = e^{-u}.$$

It is not a free phantom particle.

$g_{\mu\nu}^{\text{eff}}$

is the effective metric seen by low-energy matter modes.

$N, \quad N^i, \quad h_{ij}$

are the ADM lapse, shift and spatial metric variables in the low-energy target.

h_{ij}^{TT}

is the transverse-traceless tensor/shear radiation sector.

$u_{\mathcal{M}}^{\mu}$

denotes a would-be independent local medium-frame field. It must not appear as an independent physical field in the tested RMG core.

$Q_{AB}^{\mu\nu}$

is the highest-derivative low-energy kinetic response matrix for perturbations of matter and geometry modes.

$$G^{\mu\nu}$$

is the common principal metric kernel, identified with $g_{\text{eff}}^{\mu\nu}$ if the factorization gate succeeds.

$$K_{AB}$$

is an internal/species kinetic matrix. It carries allowed differences between particle families, tail shapes, polarizations, phases and internal geometries.

$$B_{AB}^{\mu\nu}$$

is the non-factorizing failure matrix. It must vanish on physical low-energy principal modes, up to gauge, internal or suppressed correction terms.

$$\Delta_{\mathcal{M}}$$

is the residual medium contribution to the low-energy Hamiltonian/constraint algebra after the ADM/EH target terms are separated. The tested core requires $\Delta_{\mathcal{M}} = 0$, or suppression outside all tested regimes.

3 Primitive Postulates

Primitive Postulate 1 (One base-medium). *All low-energy matter modes, rods, clocks, signals and geometric excitations are collective modes of one base-medium $\mathcal{M}$.*

Primitive Postulate 2 (Stable low-energy phase). *The base-medium has a stable low-energy phase in which matter modes, rods, clocks, signals and operational geometry are well-defined.*

Primitive Postulate 3 (Minimal operational node scale). *The base-medium as one ontological object has no ordinary microscopic radius. A length becomes physical only when the base-medium is distinguishable as recordable nodes. The first distinguishable node/record scale is denoted*

$$\ell_*.$$

Below ℓ_ , two nodes are not yet separate operational objects: there is no independent distance, delay, clock comparison or mass comparison. Above ℓ_* , medium dilution also rescales rods, clocks and the visible node scale. Thus ℓ_* is an operational closure scale, not the radius of a mechanical particle.*

In the present RMG-1 core, ℓ_ is identified with the observed Planck-order minimal record scale. This is not a second medium and not a species-dependent parameter; it is the universal first readable separation of the one base-medium.*

Primitive Postulate 4 (Operational Relational Covariance). *Local low-energy observables are invariant under re-embedding of the same relational medium-history:*

$$\mathcal{W} \sim X_F(\mathbb{R} \times \Sigma) \sim X_{F'}(\mathbb{R} \times \Sigma).$$

Thus coordinates, slicings and internal clock choices are descriptions, not extra physical structures. In particular, no independent local preferred-frame field $u_{\mathcal{M}}^\mu$ appears in the tested core.

Primitive Postulate 5 (Positive physical energy). *Physical excitations of $\mathcal{M}$ have Hamiltonian bounded below:*

$$H - H_{\text{vac}} \geq 0.$$

There is no fundamental phantom degree of freedom. The DeWitt trace sign is not a propagating phantom because the volume direction is constrained.

Primitive Postulate 6 (Local effective hierarchy). *The low-energy theory is local and organized by a derivative or scale expansion:*

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{core}} + \ell^2 \mathcal{L}_4 + \ell^4 \mathcal{L}_6 + \dots.$$

Corrections may exist, but they must be suppressed or forbidden in tested local regimes.

4 Operational Bridge

This section is not the microscopic derivation. It is the minimal bridge needed to recover the weak gravitational sector.

The universal factor $s(x)$ is not primitive. It is a bridge variable whose deeper origin should be the universal principal symbol of all low-energy matter modes.

Working Assumption 1 (Universal operational factor). *In the slowly varying low-energy regime, all stable matter modes share one local operational factor:*

$$s = s(x).$$

Equivalently, all matter modes see one effective metric $g_{\mu\nu}^{\text{eff}}$.

Status:

bridge assumption; derived below inside the coherent stable-record completion, full universal closure still open

Working Assumption 2 (Reciprocal clock/rod response). *Local clocks and rods respond as:*

$$d\tau = s dt, \quad d\ell = s^{-1} dx.$$

Result 1 (Square signal law). *If the local physical signal speed is $c = d\ell/d\tau$, then:*

$$c = \frac{s^{-1} dx}{s dt} = s^{-2} \frac{dx}{dt}.$$

Therefore:

$$\boxed{\frac{c_{\text{coord}}}{c} = s^2.}$$

Result 2 (Coherent endpoint-counting route to the square law). *The reciprocal clock/rod bridge is not produced by a longer courier path. A longer path at fixed carrier speed gives only a linear delay. A square response is obtained when the stable record sector is controlled by one coherent base-medium amplitude*

$$\boxed{\rho := |\Phi_{\mathcal{M}}| = \sqrt{n}}$$

and the local record coefficients scale as

$$\boxed{C_t : C_0 : C_x = 1 : \rho : \rho^2.}$$

Here C_t is the common-rhythm time coefficient, C_0 is the local stable gap coefficient and C_x is the spatial link-stiffness coefficient. Define

$$s := \rho^{1/2} = n^{1/4}.$$

Then

$$C_t : C_0 : C_x = 1 : s^2 : s^4.$$

For a stable quadratic record mode,

$$\omega_0 \propto \sqrt{\frac{C_0}{C_t}} = s, \quad \ell_{\text{coord}} \propto \sqrt{\frac{C_x}{C_0}} = s, \quad c_{\text{coord}} \propto \sqrt{\frac{C_x}{C_t}} = s^2 c.$$

Thus the coherent endpoint-counting completion derives

$$d\tau = s dt, \quad d\ell = s^{-1} dx, \quad \frac{c_{\text{coord}}}{c} = s^2.$$

The exponent origin is:

$$\text{site/local gap} \sim \rho, \quad \text{link/spatial stiffness} \sim \rho_a \rho_b \rightarrow \rho^2.$$

Working Assumption 3 (Additive compression and multiplicative response). *Weak compression potentials add:*

$$u_{12} = u_1 + u_2.$$

Process factors multiply:

$$s_{12} = s_1 s_2.$$

Result 3 (Exponential layering). *The continuous solution with weak-field normalization is:*

$$s = e^{-u}.$$

Result 4 (ADM lapse identification). *For a normal ADM observer with zero shift displacement,*

$$d\tau = N dt.$$

The RMG bridge clock response gives:

$$d\tau = s dt.$$

Therefore, inside the operational bridge,

$$N = s = e^{-u}.$$

The remaining microscopic burden is to derive the universal factor s and the exponential layering from $\mathcal{M}$.

Result 5 (Weak operational metric). *The bridge metric is:*

$$ds^2 = -e^{-2u} c^2 dt^2 + e^{2u} \delta_{ij} dx^i dx^j.$$

5 One-Kernel Factorization Gate

The universal matter metric must ultimately come from a common highest-derivative kinetic kernel.

$$\boxed{\mathcal{R}}$$

is the record sector: low-energy modes that can participate in operational measurement as rods, clocks, detector components, memory/record components, exchanged signals or stable external matter excitations.

$$\boxed{\mathcal{I}}$$

is the internal sector: confined internal modes, gauge directions, constraint variables, off-shell auxiliary modes, short-range internal responses and high-energy modes outside the tested local regime.

Internal modes may contribute to mass, binding energy, charge, spin, color, tail shape, effective couplings and total energy. They must not define a separate external spacetime cone in the tested local core.

For low-energy perturbations $\delta\Phi^A$, define the principal kinetic response:

$$Q_{AB}^{\mu\nu}.$$

For a covector k_μ , define the principal symbol:

$$\boxed{P_{AB}(k) = Q_{AB}^{\mu\nu} k_\mu k_\nu.}$$

Let $\Pi_{\mathcal{R}}$ project onto the physical record subspace after gauge and constraint directions are removed.

Open Theorem 1 (Factorization gate). *Derive from the protected base-medium phase:*

$$\boxed{Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB}.}$$

Then all physical low-energy modes share one principal metric:

$$\boxed{G^{\mu\nu} = g_{\text{eff}}^{\mu\nu}.}$$

Common characteristic cones alone are not enough. The physical principal symbol must also have no unsuppressed species or polarization birefringence. One base-medium alone is also not enough: one medium can support many internal or confined modes. The RMG claim is narrower:

$$\boxed{\text{one stable record-forming low-energy phase} \Rightarrow \text{one operational spacetime cone for records.}}$$

Modes outside the record sector may have different internal dispersions, but they must not define separate external rods, clocks, signals or spacetime metrics. The sharpened pass condition is:

$$\boxed{\Pi_{\mathcal{R}} P(k) \Pi_{\mathcal{R}} = (G^{\mu\nu} k_\mu k_\nu) K_{\mathcal{R}}.}$$

Allowed remainders are gauge directions, constraint variables, internal algebra, lower-derivative terms or suppressed corrections. The local core fails if the physical principal symbol contains:

$$\boxed{P_{AB}^{\text{phys}}(k) = (G^{\mu\nu} k_\mu k_\nu) K_{AB} + B_{AB}(k), \quad B_{AB}(k) \neq 0,}$$

where B_{AB} produces an observable species-private cone, polarization splitting, preferred-frame tensor or clock-dependent redshift.

Open Theorem 2 (Record-sector no-birefringence principle). *All stable modes that can form local records, rods, clocks and exchanged signals must belong to one operational record sector. In that sector:*

$$B_{AB}^{\text{phys}}(k) = 0$$

for unsuppressed principal spacetime response. Internal differences may remain in K_{AB} , masses, charges, spin, color, topology, phase and tail geometry; they must not appear as separate external spacetime cones. This is a protection principle in RMG-1 and still requires derivation from $\mathcal{M}$.

Open Theorem 3 (Record-phase route to factorization). *Derive the chain:*

$$\boxed{\text{stable record phase} \Rightarrow \text{common operational records} \Rightarrow \text{no observable species-private spacetime cone} \Rightarrow \text{no physical}}$$

This is the central local-core theorem still required from $\mathcal{M}$.

Open Theorem 4 (One-tensor factorization lemma). *For any candidate common record tensor $G^{\mu\nu}$, decompose the principal kinetic matrix as:*

$$Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB} + B_{AB}^{\mu\nu}.$$

If the stable record phase contains only one observable rank-two spacetime principal tensor, then the non-factorizing part $B_{AB}^{\mu\nu}$ has no unsuppressed physical projection on external record-forming modes:

$$\Pi_{\mathcal{R}} B_{AB}^{\mu\nu} \Pi_{\mathcal{R}} = 0$$

up to gauge, confined, gapped, damped, unstable or high-energy suppressed pieces. Otherwise $B_{AB}^{\mu\nu}$ would define a second observable cone, anisotropy, preferred-frame tensor or species-private clock response. This is an open lemma: RMG must still derive the one-observable-tensor premise from the base medium.

Gate 1 (Observable tensor basis gate). *In the tested local record core, the physical principal symbol must contain no unsuppressed observable rank-two tensor other than the common record tensor $G^{\mu\nu}$. Equivalently, any decomposition*

$$Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB} + B_{AB}^{\mu\nu}$$

must satisfy

$$\Pi_{\mathcal{R}} B_{AB}^{\mu\nu} \Pi_{\mathcal{R}} = 0$$

at leading principal order, except for gauge, constraint, internal/confined, gapped, damped, unstable, high-energy suppressed or explicitly large-scale correction sectors. Forbidden local-core pieces include unsuppressed

$$\boxed{U^\mu U^\nu, \quad v^\mu v^\nu, \quad \sigma^{\mu\nu}, \quad \omega^\mu{}_\alpha \omega^{\alpha\nu}, \quad \nabla^\mu u \nabla^\nu u, \quad S_A^{\mu\nu}.$$

If any such tensor survives in physical record propagation, RMG predicts a preferred frame, anisotropic cone, species-private metric or birefringence. This gate is open from $\mathcal{M}$.

Open Theorem 5 (Master gate: no-birefringent record carrier). *The master local-core gate is:*

$$\boxed{\text{all low-energy record-forming tails project onto one common carrier}}$$

so that the physical record-sector principal symbol has no unsuppressed species or polarization birefringence:

$$P_{AB}^{\text{phys}}(k) = (G^{\mu\nu} k_\mu k_\nu) K_{AB}.$$

If this gate closes, then one operational cone, one universal clock factor $s(x)$, the ADM lapse bridge $N = s = e^{-u}$, the total-energy source gate and the compact internal-energy/no-external-cone split become linked consequences of the same record-sector protection.

Here “one common carrier” means one shared base-medium vibration channel, not one elementary scalar field and not one tail substance per particle. Different sources may imprint different local patterns on the same medium; the tested local core allows only one external record carrier/cone.

Open Theorem 6 (Stable-tail common-carrier route). *For every stable record-forming oscillon or matter mode A , decompose its long tail as:*

$$T_A = \Pi_{\text{common}} T_A + \Pi_{\text{internal}} T_A.$$

The theorem target is:

$$\text{principal}(\Pi_{\text{common}} T_A) = G^{\mu\nu} k_\mu k_\nu \quad \text{for all record-forming } A,$$

while $\Pi_{\text{internal}} T_A$ may carry source-specific frequency, phase, spin, charge, topology, multipoles and short-range structure, but must not define a separate external spacetime cone. T_A is a source imprint in the same base-medium vibration, not a separate long-range field assigned to species A . Modes whose tails cannot project onto the common carrier are internal, confined, gapped, constrained or non-record-forming.

Open Theorem 7 (Universal operational factor from one cone). *For a clock family A , write:*

$$d\tau_A = s_A(x) dt.$$

If the one-kernel factorization gate holds, then all record-forming clock families read the same spacetime kernel $G^{\mu\nu}$. Hence:

$$s_A(x) = C_A s(x) [1 + \epsilon_A(x)].$$

Here C_A is a constant internal clock calibration, while $\epsilon_A(x)$ is the dangerous species-dependent environmental response. The tested core requires:

$$\epsilon_A(x) = 0$$

or suppression outside tested regimes. After constant calibration,

$$s_A(x) = s_B(x) = s(x).$$

Equivalently,

$$\nabla_\mu \ln \frac{s_A}{s_B} = 0$$

up to suppressed corrections. If this fails, the failure is the clock-level version of a nonzero $B_{AB}^{\mu\nu}$.

This does not mean all particle tail imprints are identical. Different oscillons may excite the same base-medium with different local frequencies, phases, polarizations and internal geometries. The factorization requirement says only that the external record carrier of their long-range tails is the same base-medium vibration channel:

$$\boxed{\text{diverse local imprints, one base-medium, one operational principal cone.}}$$

The possible failure is:

$$\boxed{Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB} + B_{AB}^{\mu\nu}, \quad B_{AB}^{\mu\nu} \neq 0.}$$

In the tested core, $B_{AB}^{\mu\nu}$ must not contain an independent preferred-frame tensor, species-private metric, or geometry-sector cone different from the matter cone.

Status:

$$\boxed{\text{open theorem / core gate, not yet derived from } \mathcal{M}.}$$

6 Tail Superposition And Projection Gate

Different oscillons may produce different local tail imprints in the same base-medium. Gravity does not read the full microscopic imprint. It reads the common pressure/refraction projection:

$$\boxed{\Pi_{\text{grav}}(T_A) = q_A \mathcal{G}.}$$

Here T_A is the source imprint associated with particle family A , $\mathcal{G}$ is the common base-medium pressure carrier, and q_A is the pressure-deficit strength. T_A must not be read as a separate long-range field or separate spacetime channel.

Open Theorem 8 (Universal gravitational projection). *For all stable low-energy record-forming matter modes, derive:*

$$\boxed{q_A \propto E_A/c^2.}$$

Here E_A means total physical energy of the source mode or composite system:

$$\boxed{E_A = E_{\text{rest}} + E_{\text{kinetic}} + E_{\text{binding}} + E_{\text{internal}}.}$$

Confined/internal modes do not need their own free long gravitational tails; they contribute through the total energy of the composite external tail.

Then many different tails source one pressure profile:

$$\boxed{u(\mathbf{x}) = \sum_i u_i(\mathbf{x}), \quad \nabla^2 u \sim -\rho_E}$$

in the weak-field regime, and the same u determines the effective geometry.

Equivalently, the far long-range pressure tail must have an energy monopole:

$$\boxed{u_A(r, \Omega) = \frac{Q_A^{(0)}}{r} + O(r^{-2}), \quad Q_A^{(0)} = \eta E_A/c^2.}$$

The theorem target is that the common pressure source density equals total physical energy density in mass units:

$$\sigma_{\text{grav}} = \alpha_E \rho_E / c^2.$$

This is expected because the long common pressure channel should couple to the conserved scalar generator of stable medium evolution, namely total energy.

The sharper source theorem target is:

$$S_u(x) := \frac{1}{N\sqrt{h}} \frac{\delta S_m}{\delta u(x)} = \rho_E(x) + \text{suppressed corrections.}$$

Here S_u is the source seen by the compression/lapse variable u . The reason is that u changes the local operational process rate, and the generator of local process-rate change is the Hamiltonian density. Thus the common pressure channel should read $\rho_E = \mathcal{H}_{\text{phys}}$, not a species-private tail detail.

Species-specific tail structure may remain in K_{AB} or internal sectors:

$$\text{spin, charge, color, phase, polarization, topology.}$$

It must not create a second long-range spacetime carrier.

Status:

$$\text{open theorem / projection gate, not yet derived from oscillon dynamics.}$$

Open Theorem 9 (Lapse/compression source theorem). *Derive from the base-medium effective action that the universal source of the compression/lapse variable u is the physical Hamiltonian density:*

$$\frac{1}{N\sqrt{h}} \frac{\delta S_m}{\delta u(x)} = \rho_E(x)$$

up to gauge terms, internal divergences or suppressed corrections. If an unsuppressed species-private source term remains, the equivalence-principle source gate fails.

Open Theorem 10 (Matter-coupled closure and source universality). *Vacuum ADM closure is not enough. The matter-coupled constraints must close:*

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_{\text{grav}} + \mathcal{H}_{\text{matter}} \approx 0, \quad \mathcal{H}_{i,\text{tot}} = \mathcal{H}_{i,\text{grav}} + \mathcal{H}_{i,\text{matter}} \approx 0.$$

The closure target is:

$$\{H_{\text{tot}}[N], H_{\text{tot}}[M]\} = D_{\text{tot}}[h^{ij}(N\partial_j M - M\partial_j N)].$$

This requires stable matter families to use the same record metric and the same total-energy source:

$$S_{\text{matter}} = S_{\text{matter}}[g, \Psi], \quad S_u = \rho_E.$$

No unsuppressed species-private gravitational source may remain:

$$\rho_{\text{grav}} \neq \rho_E + \sigma_{\text{species}} \quad \text{with } \sigma_{\text{species}} \text{ physical and unsuppressed.}$$

Internal/confined modes, including binding and gluon-like internal energy, must contribute to E_{total} while not defining an independent external spacetime carrier. This gate is open from $\mathcal{M}$.

For the ADM matter action

$$S_m = \int dt d^3x \left(\pi_A \dot{\psi}_A - N \mathcal{H}_m - N^i \mathcal{H}_i \right), \quad N = e^{-u},$$

variation at fixed h_{ij} gives

$$\delta_u S_m = \int dt d^3x N \mathcal{H}_m \delta u, \quad \frac{1}{N\sqrt{h}} \frac{\delta S_m}{\delta u} = \rho_E.$$

Thus the low-energy ADM bridge already identifies the u -source with the Hamiltonian density. The remaining RMG burden is to derive $N = e^{-u}$ and the universal ADM bridge from the base-medium.

7 Delta-M Candidate Theorem

Open Theorem 11 (Conditional Delta-M theorem). *If the protected low-energy phase has:*

one stable operational record phase with a common carrier/principal cone,

exact Operational Relational Covariance,

$$Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB},$$

and a local two-derivative low-energy hierarchy, then the tested core permits no unsuppressed independent medium residue:

$$\Delta_{\mathcal{M}} = 0$$

up to suppressed high-energy or large-scale correction windows.

Status:

conditional theorem; its premises are still open from $\mathcal{M}$.

8 Static Weak-Field Sector

Working Assumption 4 (Hamiltonian source). *In the weak static regime, the compression/lapse constraint is sourced by total physical energy density ρ_E :*

$$\rho_E(\mathbf{x}) \geq 0.$$

The inertial mass of a localized source is:

$$M = \frac{1}{c^2} \int d^3x \rho_E(\mathbf{x}).$$

Working Assumption 5 (Base-node calibration of G). *The source coefficient in the weak static sector is fixed by the minimal operational node scale:*

$$G_{\text{RMG}} = \frac{c^3 \ell_*^2}{\hbar}.$$

Equivalently, the gravitational stiffness is

$$\boxed{\frac{c^4}{4\pi G_{\text{RMG}}} = \frac{\hbar c}{4\pi \ell_*^2}}.$$

When the universal operational node scale is identified with the observed Planck-order length, G_{RMG} equals the measured Newton constant. The remaining deeper task is not to choose a smaller length, but to derive ℓ_* itself from the stable base-medium record dynamics.

Below, G denotes this calibrated G_{RMG} unless explicitly stated.
Use the static energy functional:

$$\mathcal{E}[u] = \frac{c^4}{8\pi G_{\text{RMG}}} \int d^3x |\nabla u|^2 - \int d^3x \rho_E u.$$

Result 6 (Static source equation). *Varying u gives:*

$$\boxed{\nabla^2 u = -\frac{4\pi G_{\text{RMG}}}{c^4} \rho_E}.$$

With $u(\infty) = 0$:

$$u(\mathbf{x}) = \frac{G_{\text{RMG}}}{c^4} \int d^3x' \frac{\rho_E(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}.$$

Far from the source:

$$\boxed{u(r) = \frac{G_{\text{RMG}} M}{rc^2} + O(r^{-2})}.$$

Thus the gravitational charge is inertial mass because the source is total energy.

9 PPN Weak-Field Check

Using:

$$U = c^2 u, \quad U(r) = \frac{G_{\text{RMG}} M}{r},$$

the metric gives:

$$\begin{aligned} g_{00} &= -e^{-2u} = -1 + 2u - 2u^2 + O(u^3), \\ g_{ij} &= e^{2u} \delta_{ij} = (1 + 2u + O(u^2)) \delta_{ij}. \end{aligned}$$

Therefore:

$$\boxed{\gamma = 1, \quad \beta = 1}.$$

Status:

$$\boxed{\text{derived in the static weak-field bridge sector.}}$$

10 Constraint And Radiation Targets

The low-energy dynamic description must be expressible in ADM variables:

$$ds^2 = -N^2 c^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt).$$

Weak split:

$$\begin{aligned} N &= 1 - u + O(2), \\ h_{ij} &= (1 + 2u)\delta_{ij} + h_{ij}^{TT} + O(2), \\ N_i &= \mathcal{A}_i + O(2). \end{aligned}$$

Low-Energy Target 1 (Scalar and vector sectors). *The scalar/compression variable and the vector/flow variable must be constraint or near-zone variables at low energy, not independent vacuum radiation modes.*

The absence of scalar/vector radiation is protected only if lapse and shift remain non-propagating constraint variables.

Target form:

$$\begin{aligned} \nabla^2 u &= -4\pi G\rho/c^2, \\ \nabla^2 \mathcal{A}_i &= -\kappa_V G\mathcal{P}_i^T/c^3. \end{aligned}$$

Low-Energy Target 2 (Tensor/shear radiation). *The only propagating vacuum gravitational modes at low energy must be:*

$$h_+, \quad h_\times.$$

They must have positive energy and propagation speed:

$$c_{\text{GW}} = c.$$

Target wave equation:

$$\left(\frac{1}{c^2}\partial_t^2 - \nabla^2\right) h_{ij}^{TT} = \kappa_T G T_{ij}^{TT}/c^4.$$

Vacuum:

$$\left(\frac{1}{c^2}\partial_t^2 - \nabla^2\right) h_{ij}^{TT} = 0.$$

Gate 2 (GW detection: common-mode cancellation versus shear signal). *The scalar/compression channel u is not automatically an interferometer signal. A uniform or adiabatic compression/breathing of the local record system rescales the apparatus, clock, ruler and light standard together. Therefore the local Michelson-like difference channel must project out the pure common-mode part:*

$$\boxed{\Pi_{\text{det}} \delta g_{\text{common}} = 0.}$$

The detector-visible gravitational-wave channel must instead be a differential record deformation:

$$\boxed{\Pi_{\text{det}} \delta g_{\text{rec}} = \Pi_{TT} \delta g_{\text{rec}} + \Pi_{\text{shear/defect}} \delta u_{\text{noncommon}}.}$$

The first term is the protected tensor/shear sector h_{ij}^{TT} . The second term is allowed only as a controlled conversion of breathing/compression into detector-visible shear, tidal or symmetry-defect response. It must not survive as an unsuppressed local scalar radiation mode.

This gate is the RMG version of the Michelson–Morley lesson: a local apparatus cannot detect a pure uniform rescaling of the same medium from which it is made, but it can detect anisotropic or differential record deformation. Thus RMG must explain observed gravitational-wave signals through protected TT/shear response, or through a suppressed noncommon conversion channel, never through a pure common-mode scalar breathing alone.

Open Theorem 12 (Constraint protection route). *Derive from the base-medium that the local low-energy variables enter in ADM-like form:*

$$S_{\text{local}} = \int dt d^3x \left(\pi^{ij} \dot{h}_{ij} - N\mathcal{H} - N^i \mathcal{H}_i \right)$$

with no independent low-energy kinetic terms

$$\dot{N}^2, \quad \dot{N}_i \dot{N}^i, \quad \dot{u}^2, \quad \dot{A}_i \dot{A}^i$$

for physical scalar/vector gravitational variables. If N and N^i impose first-class Hamiltonian and momentum constraints which close under time evolution, then u and local flow/shift variables are constraint/gauge or near-zone variables, not independent radiation. The vacuum gravitational record sector then has only

$$h_+, \quad h_\times$$

as physical propagating modes. This is the local mechanism behind flow decoupling and the preferred-frame null target.

11 ADM/EH Fixed Point

Low-Energy Target 3 (Safe nonlinear completion). *The safe nonlinear low-energy fixed point is:*

$$S_{\text{low}} = \frac{c^4}{16\pi G} \int d^4x \sqrt{-g} R + S_{\text{matter}}[g, \Psi].$$

ADM form:

$$S_{\text{ADM}} = \int dt d^3x \left(\pi^{ij} \dot{h}_{ij} - N\mathcal{H} - N^i \mathcal{H}_i \right).$$

Hamiltonian and momentum constraints:

$$\mathcal{H} \approx 0, \quad \mathcal{H}_i \approx 0.$$

The target constraint algebra includes:

$$\{H[N], H[M]\} = D[h^{ij}(N\partial_j M - M\partial_j N)].$$

Operational Relational Covariance makes ADM closure plausible, but it does not replace the proof.

Open Theorem 13 (Main closure theorem). *Derive from the base-medium:*

$$\mathcal{M} \Rightarrow \text{universal principal symbol} \Rightarrow \text{ADM constraint closure} \Rightarrow S_{\text{EH}} + S_{\text{matter}}.$$

Open Theorem 14 (Hypersurface-deformation closure filter). *The local Hamiltonian constraint must pass the ADM deformation-algebra filter. For the two-derivative ansatz*

$$\mathcal{H} = \frac{1}{\sqrt{h}} (\pi^{ij} \pi_{ij} - \lambda \pi^2) - \sqrt{h} (\alpha R - 2\Lambda) + \Delta_{\mathcal{M}},$$

the target bracket is:

$$\{H[N], H[M]\} = D[h^{ij} (N \partial_j M - M \partial_j N)].$$

This filter requires, in the protected local core:

$$\lambda = \frac{1}{2}, \quad \Delta_{\mathcal{M}} = 0 \quad \text{or constrained/gauge/suppressed,}$$

and no unsuppressed higher-derivative curvature terms at the tested two-derivative fixed point. If this closure filter is derived from $\mathcal{M}$, then the trace/volume sign in the DeWitt supermetric is a constraint direction, not a propagating phantom scalar, and the safe low-energy fixed point is $S_{\text{EH}} + S_{\text{matter}}$ up to suppressed corrections.

12 Local Gravity Closure Package

The many open local-core gates reduce to three root theorems.

Open Theorem 15 (Root A: one external record carrier). *Derive from the base-medium stable record phase:*

$$\text{stable external record-forming mode} \Rightarrow \text{one common external long-tail carrier channel.}$$

All non-common branches must be internal, confined, gapped, damped, constrained, gauge or unstable. If Root A closes, it gives one operational principal cone, no low-energy birefringence and the factorization

$$Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB}$$

on physical record modes.

Open Theorem 16 (Root A spectral/resonance closure). *Root A must be proven by a spectral statement around the stable record phase $\mathcal{M}_0$. Linearize the base-medium dynamics:*

$$\delta \mathcal{E}[\mathcal{M}_0] = \mathcal{L}_{\mathcal{M}_0} \delta \Phi + \dots$$

The required split is:

$$\text{Spec}(\mathcal{L}_{\mathcal{M}_0}) = \mathcal{B}_{\text{rec}} \cup \mathcal{B}_{\text{int}}, \quad N_{\text{cone}}(\mathcal{B}_{\text{rec}}^{\text{gapless}}) = 1$$

as one external record carrier channel. This is a one-common-carrier condition, not a one-scalar-mode condition. The common channel may contain shared polarization, shear or tensor patterns, but it must not split into species-private external carriers. Stable record-forming tail imprints must phase-lock to the base-medium rhythm ν_0 or its allowed harmonic channel:

$$\Pi_{\text{rec}} T_A^{\text{long}} = Z_A C + \text{internal/lower-order details}, \quad \text{principal}(C) = G^{\mu\nu} k_\mu k_\nu.$$

All other branches must be internal, confined, gapped, damped, constrained, gauge, unstable, high-energy suppressed or large-scale-only. This is the microscopic calculation still needed for Root A.

Open Theorem 17 (Root B: record-embedding ADM closure). *Derive that changes of slicing, labels and record parametrization are gauge redundancies of the same physical record history. These redundancies must generate first-class constraints:*

$$\boxed{\mathcal{H} \approx 0, \quad \mathcal{H}_i \approx 0,}$$

with the ADM hypersurface-deformation algebra. If Root B closes, local scalar/vector gravitational sectors are constrained, flow is shift-like in the tested core, only $h_+, h_\times$ propagate, $\lambda = 1/2$ is selected and the two-derivative fixed point is S_{EH} .

Open Theorem 18 (Root B embedding/symplectic closure). *The precise Root B burden is to show that the physical object is a relational record history $W_{\mathcal{R}}$, while a foliation/embedding X_F is only a description of that history:*

$$\boxed{X_F \sim X'_F, \quad \Gamma_{\text{phys}} = \Gamma_{\mathcal{R}}/\mathcal{G}_{\text{emb}}.}$$

This equivalence must be a true gauge degeneracy of the record symplectic structure:

$$\boxed{\iota_{\delta\xi}\Omega_{\mathcal{R}} \approx 0.}$$

Then the embedding generators must take the ADM form

$$\boxed{G[\xi] = D[N^i] + H[N], \quad D[N^i] = \int d^3x N^i \mathcal{H}_i, \quad H[N] = \int d^3x N \mathcal{H},}$$

with

$$\boxed{\mathcal{H}_i \approx 0, \quad \mathcal{H} \approx 0.}$$

The composition of two embedding changes must be the hypersurface-deformation algebra:

$$\boxed{\{D[N^i], D[M^i]\} = D[[N, M]^i],}$$

$$\boxed{\{H[N], D[M^i]\} = H[\mathcal{L}_M N],}$$

$$\boxed{\{H[N], H[M]\} = D[h^{ij}(N\partial_j M - M\partial_j N)],}$$

where the structure function h^{ij} is the same inverse spatial metric seen by the record sector. Lapse and shift are then embedding multipliers, not extra physical medium particles:

$$\boxed{S_{\text{loc}} = \int dt d^3x \left(\pi^{ij} \dot{h}_{ij} - N \mathcal{H} - N^i \mathcal{H}_i \right), \quad \dot{N}^2 = \dot{N}_i \dot{N}^i = 0.}$$

If this theorem closes, the scalar and vector gravitational sectors are constraints/near-zone book-keeping, local flow is shift-like, only two tensor modes propagate, the DeWitt value $\lambda = 1/2$ is selected and the two-derivative local fixed point is S_{EH} .

Open Theorem 19 (Root B finite sufficiency checklist). *Root B is reduced to five sufficient conditions on the low-energy record sector:*

- | |
|--|
| B1 no observable absolute record time,
B2 one record geometry h_{ij} ,
B3 embedding changes are presymplectic null directions,
B4 null directions have local generators $D[N^i], H[N]$,
B5 generator composition is anomaly-free. |
|--|

The algebraic danger is a residual anomaly in the normal-normal bracket:

$$\{H[N], H[M]\} = D[h^{ij}(N\partial_j M - M\partial_j N)] + A[N, M].$$

Root B requires

$$A[N, M] = 0$$

on the local tested record sector. If $A \neq 0$, then at least one hidden preferred-frame, scalar, vector or non-record tensor structure has survived. Equivalently, lapse/shift or local flow has become physical rather than embedding bookkeeping. The action-level check is:

$$\dot{N}^2, \quad \dot{N}_i \dot{N}^i, \quad \text{unsuppressed independent scalar/vector kinetic terms} \quad \text{are absent.}$$

Thus Root B is not an aesthetic GR assumption. It is the finite test:

$$B1 + B2 + B3 + B4 + B5 \Rightarrow \text{ADM local closure.}$$

Open Theorem 20 (Root B intuition audit). *The base-medium intuition supports Root B in a graded way:*

condition	intuition audit status
B1	strong support: time/measurement are operational and relational
B2	partial: requires Root A one-carrier closure
B3	open: requires a record presymplectic form $\Omega_{\mathcal{R}}$
B4	open: requires local moment-map generators D, H
B5	open: requires anomaly cancellation $A[N, M] = 0$

The intuitive file says that a single undivided base medium has no inner ruler, delay or comparison until it self-differentiates into distinguishable record-forming nodes. This strongly supports the absence of an observable absolute record time. It does not by itself prove the canonical statement

$$\iota_{\delta_\xi} \Omega_{\mathcal{R}} \approx 0 \quad \Rightarrow \quad G[\xi] = D[N^i] + H[N].$$

Therefore the next exact Root B burden is to construct the low-energy record phase space $\Gamma_{\mathcal{R}}$, its presymplectic form $\Omega_{\mathcal{R}}$, the embedding-variation vector fields δ_ξ , and the local generators D, H . The pass condition remains

$$\{H[N], H[M]\} - D[h^{ij}(N\partial_j M - M\partial_j N)] = 0$$

in the tested local record sector.

Open Theorem 21 (Record phase space and presymplectic Root B construction). *The non-circular Root B construction starts from the covariant record phase space*

$$\Gamma_{\mathcal{R}} = \{\Phi_{\mathcal{R}} \text{ solving the record equations}\} / \mathcal{G}_{\text{int}}, \quad \Phi_{\mathcal{R}} = (g_{\text{rec}}, \Psi_{\text{rec}}),$$

where $\mathcal{G}_{\text{int}}$ removes internal/non-record redundancies but does not yet quotient embedding freedom. Let the low-energy record action satisfy

$$\delta L_{\mathcal{R}} = E_{\mathcal{R}} \delta \Phi_{\mathcal{R}} + d\theta_{\mathcal{R}}(\delta \Phi_{\mathcal{R}}).$$

Define the covariant presymplectic current

$$\omega_{\mathcal{R}}(\delta_1 \Phi, \delta_2 \Phi) = \delta_1 \theta_{\mathcal{R}}(\delta_2 \Phi) - \delta_2 \theta_{\mathcal{R}}(\delta_1 \Phi)$$

and the slice form

$$\Omega_{\mathcal{R}, \Sigma} = \int_{\Sigma} \omega_{\mathcal{R}}.$$

On shell and without boundary flux, $d\omega_{\mathcal{R}} = 0$, so

$$\Omega_{\mathcal{R}, \Sigma} = \Omega_{\mathcal{R}, \Sigma'}.$$

This is the formal version of one record history described by different slices. An embedding deformation acts as

$$\delta_{\xi} \Phi_{\mathcal{R}} = \mathcal{L}_{\xi} \Phi_{\mathcal{R}} \quad \text{up to internal gauge compensation.}$$

Root B requires this direction to be presymplectically null:

$$\Omega_{\mathcal{R}}(\delta \Phi, \delta_{\xi} \Phi) \approx 0.$$

Equivalently, the Noether current

$$J_{\xi} = \theta_{\mathcal{R}}(\mathcal{L}_{\xi} \Phi_{\mathcal{R}}) - \xi \cdot L_{\mathcal{R}}$$

must split as

$$J_{\xi} = C_{\xi} + dQ_{\xi}, \quad \int_{\Sigma} C_{\xi} = H[N] + D[N^i], \quad \xi = Nn + N^i e_i.$$

The pass condition is therefore

$$\Omega_{\mathcal{R}}(\delta \Phi, \mathcal{L}_{\xi} \Phi) = \delta(H[N] + D[N^i])$$

with $H \approx 0$ and $D \approx 0$ in the tested local record sector. Any bulk remainder that is not a constraint or boundary charge is the Root B preferred-frame/scalar/vector anomaly.

Open Theorem 22 (Noether split and background-free record-action gate). *The Noether split required by Root B follows if the local tested record action is background-free:*

$$L_{\mathcal{R}}^{\text{local}} = L_{\mathcal{R}}[g_{\text{rec}}, \Psi_{\text{rec}}]$$

up to constraints, gauge choices, boundary terms, integrated-out internal contributions, high-energy suppressed corrections and large-scale-only flow/MOND terms. In that case an embedding deformation is a record diffeomorphism:

$$\delta_{\xi} L_{\mathcal{R}} = \mathcal{L}_{\xi} L_{\mathcal{R}} = d(\xi \cdot L_{\mathcal{R}}),$$

and the Noether current can be written as

$$J_{\xi} = C_{\xi} + dQ_{\xi}, \quad C_{\xi} = N\mathcal{H} + N^i \mathcal{H}_i.$$

The forbidden local failure is a non-record background structure B :

$$\boxed{L_{\mathcal{R}}^{\text{local}} = L_{\mathcal{R}}[\Phi_{\mathcal{R}}; B].}$$

If B is not a gauge, constraint, boundary, integrated-out, suppressed or large-scale-only object, then the embedding variation leaves a bulk anomaly:

$$\boxed{\mathcal{A}_{\xi} \sim \frac{\delta L_{\mathcal{R}}}{\delta B} \mathcal{L}_{\xi} B.}$$

The local-core forbidden B 's are:

$$\boxed{T_{\mathcal{M}}, \quad U_{\mathcal{M}}^{\mu}, \quad v^{\mu}v^{\nu}, \quad \text{species-private metric/clock}, \quad \dot{N}^2, \quad \dot{N}_i\dot{N}^i, \quad \text{extra record cone.}}$$

Every bulk remainder in J_{ξ} must therefore be classified as exactly one of:

$$\boxed{\text{constraint/gauge/boundary/internal/suppressed/large-scale-only/anomaly.}}$$

Root B requires the final class to vanish in the local tested sector:

$$\boxed{\mathcal{A}_{\xi}^{\text{local}} = 0 \quad \Longleftrightarrow \quad A[N, M] = 0.}$$

Open Theorem 23 (Background-free two-derivative record-action fixed point). *If Root A supplies one record metric g_{rec} , Root B removes all local non-record background structures, and the tested local regime admits a two-derivative effective hierarchy, then the local record action has the form*

$$\boxed{L_{\mathcal{R}}^{\text{local}} = \sqrt{-g_{\text{rec}}} [\alpha R[g_{\text{rec}}] - 2\Lambda + L_{\text{m}}(g_{\text{rec}}, \Psi_{\text{rec}})] + dB + O(\partial^4/M_*^2).}$$

At the two-derivative fixed point this is

$$\boxed{S_{\text{EH}} + S_{\text{matter}}}$$

up to normalization, cosmological term, boundary terms and suppressed higher-derivative corrections. The forbidden local background leaks are:

$$\boxed{U_{\mathcal{M}}^{\mu} U_{\mathcal{M}}^{\nu} \nabla_{\mu} \Psi \nabla_{\nu} \Psi, \quad v^{\mu}v^{\nu} R_{\mu\nu}, \quad \dot{N}^2, \quad \dot{N}_i\dot{N}^i, \quad g_A^{\mu\nu} \partial_{\mu} \Psi_A \partial_{\nu} \Psi_A, \quad C^{\mu\nu} k_{\mu} k_{\nu} \not\propto g_{\text{rec}}^{\mu\nu} k_{\mu} k_{\nu}.}$$

Each such term must be absent, constrained, gauge, boundary-only, integrated-out/internal, high-energy suppressed or large-scale-only. Otherwise the two-derivative fixed point is not GR and Root B fails in the local tested core. Thus the local closure chain is

$$\boxed{\text{one record metric} + \text{background-free action} + \text{two-derivative hierarchy} \Rightarrow S_{\text{EH}} + S_{\text{matter}}.}$$

Open Theorem 24 (Root C: total-energy source projection). *Derive the far-tail/source theorem:*

$$\boxed{Q_A^{(0)} = \eta E_A / c^2, \quad S_u = \rho_E,}$$

where E_A includes rest, kinetic, binding and internal/confined energy. No unsuppressed species-private source may remain. If Root C closes, the equivalence-principle source gate and compact internal-energy effacement gate close.

Open Theorem 25 (Root C finite source chain). *Root C reduces to five source conditions:*

$$\begin{array}{ll}
C1 & S_m = S_m[g_{\text{rec}}, \Psi] \quad \text{universal matter action,} \\
C2 & N = e^{-u} \quad \text{or equivalent compression/lapse bridge,} \\
C3 & S_u = (N\sqrt{h})^{-1} \delta S_m / \delta u = \rho_E, \\
C4 & \nabla^2 u = -\kappa \rho_E \quad \text{far-tail Gauss law,} \\
C5 & \sigma_{\text{species}}^{\text{long, local}} = 0.
\end{array}$$

Then an isolated source obeys

$$\oint_{\partial B_R} \nabla u \cdot dS = -\kappa \int_{B_R} \rho_E d^3x, \quad u(r) = \frac{Q^{(0)}}{r} + O(r^{-2}),$$

and therefore

$$Q^{(0)} = \eta E_{\text{total}} / c^2.$$

Thus the hard part of Root C is not the far $1/r$ integral. The hard part is proving that the compression/lapse source is total physical energy for all stable record matter:

$$S_u = \rho_E$$

with no unsuppressed species-private remainder:

$$S_u \neq \rho_E + \sigma_{\text{species}} \quad \text{for physical long-range local } \sigma_{\text{species}}.$$

Species-specific structure is allowed only if it is an internal divergence, gauge, boundary multipole, high-energy suppressed, short-range/confined or already included in ρ_E . Internal, binding, gluon-like and confined energy must contribute to E_{total} while not producing an independent external cone:

$$E_{\text{total}} = E_{\text{rest}} + E_{\text{kinetic}} + E_{\text{binding}} + E_{\text{internal}}.$$

Open Theorem 26 (C3 lapse-variation source theorem). *Assume the stable record matter sector satisfies*

$$S_m = S_m[g_{\text{rec}}, \Psi], \quad N = e^{-u},$$

and has no unsuppressed explicit species-private u -coupling outside the record metric. In canonical form,

$$S_m = \int dt d^3x \left(\pi_A \dot{\psi}_A - N \mathcal{H}_m - N^i \mathcal{H}_{i,m} \right).$$

At fixed $h_{ij}, N^i, \psi_A, \pi_A,$

$$\delta N = -N \delta u, \quad \delta_u S_m = \int dt d^3x N \mathcal{H}_m \delta u.$$

Therefore

$$S_u = \frac{1}{N\sqrt{h}} \frac{\delta S_m}{\delta u} = \frac{\mathcal{H}_m}{\sqrt{h}} = \rho_E.$$

This is the precise Root C bridge: changing compression/lapse changes the local process rate, and the generator of local process-time translation is the Hamiltonian density. Internal, binding, gluon-like and confined energy are included because

$$\mathcal{H}_m = \mathcal{H}_{\text{rest}} + \mathcal{H}_{\text{kinetic}} + \mathcal{H}_{\text{binding}} + \mathcal{H}_{\text{internal}}.$$

Root C fails only if a physical long-range local remainder survives:

$$S_u = \rho_E + \sigma_{\text{species}}^{\text{long,local}}, \quad \sigma_{\text{species}}^{\text{long,local}} \neq 0.$$

Thus the C3 gate is: all stable record matter must see u only through the same record metric/lapse, except for gauge, constrained, boundary, internal, short-range or suppressed terms already accounted for in ρ_E .

Open Theorem 27 (Metric-exhaustion and no species-private u -coupling). *If the stable local matter sector is exhausted by one record metric,*

$$S_m = S_m[g_{\text{rec}}, \Psi],$$

then all allowed local u -dependence is metric/lapse dependence. Species labels may appear in masses, charges, spin, internal Hamiltonians, confined sectors, kinetic matrices, multipoles and finite-size corrections, but not as a separate long-range local term

$$\Delta S_A = \int d^4x N \sqrt{h} F_A(u, \Psi_A).$$

If such a term is physical and unsuppressed, then

$$S_u = \rho_E + \sigma_A,$$

so u becomes a species-readable scalar background rather than just the record lapse/compression variable. This is a fifth-force/source anomaly: two systems with the same total energy but different composition would source or fall differently. Therefore the local Root C gate is

$$F_A^{\text{long,local}} = 0 \quad \Longleftrightarrow \quad \sigma_{\text{species}}^{\text{long,local}} = 0,$$

unless the term is already included in ρ_E , or is gauge, constrained, boundary, internal, short-range/confined or high-energy suppressed. In the common pressure projection language,

$$\Pi_{\text{grav}}(T_A) = q_A \mathcal{G}, \quad q_A = \eta E_A / c^2$$

must hold at monopole order; species-specific tail data may survive only as spin, charge, color, phase, multipole, internal, short-range or suppressed structure.

Open Theorem 28 (Local gravity core closure). *If Root A, Root B and Root C are derived from $\mathcal{M}$, then the local RMG gravity core is closed:*

$$\text{one operational spacetime} + S_{\text{EH}} + S_{\text{matter}} + \text{universal total-energy source}.$$

The remaining sectors – strong-field compact interiors, MOND reentry, cosmology and detailed matter emergence – then become extension windows, not loose foundations of the local gravitational core.

Open Theorem 29 (Non-circular local-core proof order). *The local gravity proof must close in the following order:*

$$\boxed{\mathcal{M}_{\text{stable}} \Rightarrow \text{Root } A \Rightarrow g_{\text{rec}} \Rightarrow \text{Root } B \Rightarrow S_{\text{EH}} + S_{\text{matter}} \Rightarrow \text{Root } C \Rightarrow S_u = \rho_E.}$$

Equivalently:

$$\boxed{\begin{array}{ll} \text{Root } A & : \text{ one common external record carrier/metric,} \\ \text{Root } B & : \text{ record re-embedding gauge } \Rightarrow \text{ ADM/EH,} \\ \text{Root } C & : \text{ metric/lapse matter coupling } \Rightarrow \text{ total-energy source.} \end{array}}$$

The proof must obey the no-circularity rules:

$$\boxed{\begin{array}{l} \text{do not assume ADM to prove Root } A, \\ \text{do not assume Einstein equations to prove Root } B, \\ \text{do not assume the equivalence principle to prove Root } C, \\ \text{do not use MOND/cosmology to repair a local-core failure.} \end{array}}$$

The microscopic object still needed from the base medium is an effective record action

$$\boxed{S_{\text{eff}}^{\text{rec}}[\mathcal{M}_0]}$$

with one record metric, no local background leak, a covariant presymplectic form, universal matter metric coupling and no explicit species-private u-coupling. This is the next exact derivation target.

13 Leading-Order Relational Base Derivation

The strongest current internal result is a leading-order derivation of the tested local gravity core from a relational base-medium action candidate. It is not yet the final microscopic theory, because the exact coefficients, matter spectrum, MOND scale and strong-field branch remain to be computed.

Result 7 (Relational base action candidate). *The base-medium is represented without external rods, clocks or background space by relational variables*

$$\boxed{R_{ab}, \quad \vartheta_{ab}, \quad K_{ab}.}$$

Here R_{ab} denotes adjacency/separation records, ϑ_{ab} the common rhythm/phase records, and K_{ab} internal coupling/shape records. The candidate base action has Jacobi/constraint form

$$\boxed{S_{\mathcal{M}}^{\text{rel}} = \int d\lambda \left[\Theta_{\mathcal{M}}(R, \vartheta, K; \dot{R}, \dot{\vartheta}, \dot{K}) - \alpha C_0 - \beta^a C_a - \gamma C_\nu \right].}$$

λ is only a bookkeeping parameter, not physical time. The constraints have the intended meanings

$$\boxed{C_0 = \text{common process constraint}, \quad C_a = \text{relabel/re-embedding constraint}, \quad C_\nu = \text{one-rhythm/harmonic selection constraint}}$$

Thus the base does not begin with spacetime. Spacetime variables are allowed only as coarse-grained record variables.

Result 8 (Continuum core route). *Under smooth coarse-graining, local isotropy, one common record carrier and no physical background-label leak, the continuum record variables are*

$$\boxed{h_{ij} = h_{ij}[R, \vartheta, K], \quad \pi^{ij} = \pi^{ij}[R, \vartheta, K].}$$

Relabel/re-embedding redundancy gives the momentum constraint:

$$\boxed{C_a \longrightarrow H_i.}$$

Small relational loop defects give curvature:

$$\boxed{\Omega_{\text{loop}} - 1 \longrightarrow R^i{}_{jkl}(h), \quad V_{\text{loop}} \longrightarrow \int d^3x \sqrt{h} (R(h) - 2\Lambda).}$$

The relational kinetic term decomposes into physical shear plus constrained volume/gauge directions. In the protected continuum limit this gives the DeWitt kinetic form

$$\boxed{T_{\text{rec}} \longrightarrow \frac{1}{\sqrt{h}} \left(\pi_{ij} \pi^{ij} - \frac{1}{2} \pi^2 \right) .}$$

Therefore the leading-order local record Hamiltonian is

$$\boxed{H = \frac{1}{\sqrt{h}} \left(\pi_{ij} \pi^{ij} - \frac{1}{2} \pi^2 \right) - \sqrt{h} (R(h) - 2\Lambda) + H_{\text{matter}}.}$$

Together with H_i , this is the ADM/Einstein-Hilbert local fixed point at leading order.

Result 9 (Radiation and source consequence). *The metric perturbation splits operationally as*

$$\boxed{\delta h_{ij} = s_{ij} + \frac{1}{3} h_{ij} b + \text{gauge/constraint pieces.}}$$

The common breathing variable b is not a separate free radiation field in the tested local sector; it is hidden by common-mode self-calibration and by the Hamiltonian constraint. The observable radiation is the shear sector

$$\boxed{s_{ij} = h_{ij}^{TT},}$$

which carries two tensor modes. Matter couples to the same record metric, so the local source is total energy rather than a species-private tail charge.

Gate 3 (Leading-order status). *RMG currently has a conditional leading-order route*

$$\boxed{\mathcal{M}_{\text{rel}} \Rightarrow \text{record geometry} \Rightarrow \text{ADM/EH local core} \Rightarrow \text{two tensor GW modes} \Rightarrow \text{total-energy source.}}$$

This is the strongest present core result. It becomes a full derivation only after the exact coarse-graining map, constants G, Λ , stable spectrum, matter couplings, MOND reentry scale and strong-field branch are computed from the base-medium rather than inserted as targets.

Open Theorem 30 ($S_{\text{eff}}^{\text{rec}}$ derivation pipeline). *Start from a base-medium action/evolution functional*

$$S_{\mathcal{M}}[\Phi_{\mathcal{M}}], \quad \Phi_{\mathcal{M}} = \Phi_0 + \delta\Phi,$$

around a stable phase $\mathcal{M}_0$. Linearization gives

$$\delta^2 S_{\mathcal{M}} = \frac{1}{2} \langle \delta\Phi, \mathcal{L}_{\mathcal{M}_0} \delta\Phi \rangle.$$

The fluctuation space must split into record and internal sectors:

$$\delta\Phi = R + I, \quad R = \Pi_{\mathcal{R}} \delta\Phi, \quad I = \Pi_{\mathcal{I}} \delta\Phi,$$

with

$$\Pi_{\mathcal{R}} + \Pi_{\mathcal{I}} = 1, \quad \Pi_{\mathcal{R}} \Pi_{\mathcal{I}} = 0.$$

The record effective action is then the low-energy functional

$$e^{iS_{\text{eff}}^{\text{rec}}[R]} = \int \mathcal{D}I e^{iS_{\mathcal{M}}[R, I]}.$$

Equivalently,

$$S_{\text{eff}}^{\text{rec}}[R] = -i \log \int \mathcal{D}I e^{iS_{\mathcal{M}}[R, I]}.$$

The record-sector principal kernel must factorize:

$$S_{\mathcal{R}}^{(2)} \sim \int R_A Q_{AB}^{\mu\nu} \partial_\mu \partial_\nu R_B, \quad Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB},$$

so that one record metric is defined by $g_{\text{rec}}^{\mu\nu} \sim G^{\mu\nu}$. The internal integration may renormalize constants, add binding energy, multipoles, finite-size terms, high-energy suppressed operators or large-scale-only corrections. It must not leave local unsuppressed

$$U_{\mathcal{M}}^\mu, \quad v^\mu v^\nu, \quad \text{species-private metric,} \quad \text{explicit species-private } u\text{-coupling,} \quad \dot{N}^2, \quad \dot{N}_i \dot{N}^i, \quad \text{extra record cone.}$$

If this pipeline succeeds, the same microscopic object supplies the raw material for Root A, Root B and Root C. If it fails, the failure is localized as either non-factorization, background leak, or source anomaly.

Open Theorem 31 (Minimal base-medium input package). *To compute $S_{\text{eff}}^{\text{rec}}$, the base-medium model must provide the input package*

$$\mathcal{D}_{\mathcal{M}} = (\Phi_{\mathcal{M}}, S_{\mathcal{M}}, \mathcal{M}_0, \ell_*, \Omega_{\mathcal{M}}, \mathcal{G}_{\text{int}}, \mathcal{O}_{\text{rec}}).$$

Here $\Phi_{\mathcal{M}}$ denotes the microscopic/collective medium variables, $S_{\mathcal{M}}$ the action or evolution functional, $\mathcal{M}_0$ the stable low-energy phase, ℓ_* the minimal operational node scale, $\Omega_{\mathcal{M}}$ the symplectic or presymplectic structure, $\mathcal{G}_{\text{int}}$ the internal redundancy group, and $\mathcal{O}_{\text{rec}}$ the operational record observables. Around $\mathcal{M}_0$,

$$\Phi_{\mathcal{M}} = \Phi_0 + \delta\Phi, \quad \delta^2 S_{\mathcal{M}} = \frac{1}{2} \langle \delta\Phi, \mathcal{L}_{\mathcal{M}_0} \delta\Phi \rangle.$$

The spectral target is

$$\text{Spec}(\mathcal{L}_{\mathcal{M}_0}) = \mathcal{B}_{\text{rec}} \cup \mathcal{B}_{\text{int}}, \quad N_{\text{cone}}(\mathcal{B}_{\text{rec}}^{\text{gapless}}) = 1$$

as one external record carrier/cone, not one scalar field. The projectors must be built from observability and spectrum, not by assuming GR:

$$\Pi_{\mathcal{R}} + \Pi_{\mathcal{I}} = 1, \quad \Pi_{\mathcal{R}}\Pi_{\mathcal{I}} = 0.$$

At quadratic order, internal integration has the block form

$$S^{(2)} = \frac{1}{2} \begin{pmatrix} R & I \end{pmatrix} \begin{pmatrix} L_{RR} & L_{RI} \\ L_{IR} & L_{II} \end{pmatrix} \begin{pmatrix} R \\ I \end{pmatrix}, \quad L_{\text{eff}} = L_{RR} - L_{RI}L_{II}^{-1}L_{IR}.$$

The Schur-complement correction may renormalize the record kernel, binding energy, multipoles or suppressed operators. It must not generate an extra cone, local preferred-frame tensor, species-private metric, explicit species-private u -coupling or lapse/shift kinetic terms. Without this input package, RMG-1 has a sharp proof target but not yet a microscopic proof.

Open Theorem 32 (Candidate base-medium order-parameter ansatz). *A first calculable microscopic target is a single base-medium order parameter, not many independent fundamental substances:*

$$\Phi_{\mathcal{M}} = \Psi_{\mathcal{M}} = \sqrt{n} e^{i\theta} \chi.$$

Here n is local node-density/pressure amplitude, θ is phase or medium rhythm, and χ denotes internal shape/orientation data of the same medium. This ansatz is allowed only if χ does not become an independent long-range spacetime carrier. It must be internal, confined, gauge, constrained, suppressed, or part of record matter sharing the same carrier.

A minimal positive-energy candidate functional is

$$E_{\mathcal{M}} = \int d^3X \left[\frac{\kappa_n}{2} (\nabla n)^2 + \frac{\kappa_\theta}{2} n (\nabla \theta)^2 + V(n) + E_{\text{int}}(n, \theta, \chi, \nabla \chi) \right],$$

with a stable phase

$$V'(n_0) = 0, \quad V''(n_0) > 0.$$

Schematically,

$$S_{\mathcal{M}} = \int d\tau \left[\Theta_{\mathcal{M}}(\Phi_{\mathcal{M}}, \dot{\Phi}_{\mathcal{M}}) - E_{\mathcal{M}}[\Phi_{\mathcal{M}}] \right],$$

where $\Theta_{\mathcal{M}}$ defines the symplectic structure. The stable phase expansion is

$$n = n_0 + \delta n, \quad \theta = \nu_0 \tau + \delta \theta, \quad \chi = \chi_0 + \delta \chi.$$

The Root A spectral target is still

$$N_{\text{cone}}(\mathcal{B}_{\text{rec}}^{\text{gapless}}) = 1.$$

This means one observable external carrier/cone, not one scalar excitation. The bookkeeping parameter τ , any preferred flow, and any extra observable carrier branch must not survive as local observables in $S_{\text{eff}}^{\text{rec}}$. The long compression variable u should arise from a pressure/density deficit, for example $u \sim -\delta \ln n$, and its weak static equation must be constraint-like:

$$\nabla^2 u = -\kappa \rho_E + \text{allowed corrections},$$

not a propagating phantom scalar. This ansatz is a test model, not a proof: if it leaves a physical medium time, preferred frame or extra cone, it must be constrained or replaced.

Result 10 (Stable-record coherent amplitude completion). *The ansatz $\Phi_{\mathcal{M}} = \sqrt{n}e^{i\theta}\chi$ by itself is not yet a microscopic proof of the square law. The square law is derived if the projected long-range stable record sector has the coherent endpoint-counting completion*

$$\boxed{\rho := |\Phi_{\mathcal{M}}| = \sqrt{n}}$$

and the first-order record action

$$\boxed{S_{\text{rec}} = \int d\lambda d^3q \left[P_A \dot{R}_A - \alpha \mathcal{C}_{\text{rec}} \right],}$$

with

$$\boxed{\mathcal{C}_{\text{rec}} = \frac{1}{2} \left[K^{AB} P_A P_B + K_{AB} \left(c^2 \rho^2 \nabla_{\text{rec}} R_A \nabla_{\text{rec}} R_B + m_A^2 \rho R_A R_B \right) \right].}$$

The multiplier α is the common process multiplier, not an external time. In local common-rhythm proper gauge, $d\tau = \alpha d\lambda$, the momenta eliminate to

$$\boxed{S_{\text{rec}}^{(2)} = \frac{1}{2} \int d\tau d^3q K_{AB} \left[\partial_\tau R_A \partial_\tau R_B - c^2 \rho^2 \nabla_{\text{rec}} R_A \nabla_{\text{rec}} R_B - m_A^2 \rho R_A R_B \right].}$$

Therefore

$$\boxed{C_t = 1, \quad C_0 = \rho, \quad C_x = \rho^2.}$$

The local dispersion relation is

$$\boxed{\omega^2 = c^2 \rho^2 k^2 + m_A^2 \rho.}$$

Consequently, with $s = \rho^{1/2} = n^{1/4}$,

$$\boxed{\omega_0 \sim s, \quad \ell_{\text{coord}} \sim s, \quad c_{\text{coord}} = cs^2.}$$

This is the precise stable-record completion behind the reciprocal clock/rod response. It does not claim that the order parameter alone proves the result; the endpoint-counting record completion is the added calculable structure.

Result 11 (Endpoint-counting exponent selection). *The powers in the coherent completion have a relational node-link origin. A local gap term has one active endpoint; hence its leading coherent amplitude weight is*

$$\boxed{L_a \propto \rho_a.}$$

A spatial link has two active endpoints and must vanish if either endpoint is inactive. Endpoint symmetry and minimal coherent order then give

$$\boxed{J_{ab} \propto \rho_a \rho_b.}$$

In a smooth patch, $\rho_a, \rho_b \rightarrow \rho$, so the stable record sector has

$$\boxed{\text{gap} \sim \rho, \quad \text{link stiffness} \sim \rho^2.}$$

This selects $C_t : C_0 : C_x = 1 : \rho : \rho^2$, and therefore the square law $c_{\text{coord}}/c = s^2$ with $s = \rho^{1/2}$.

Open Theorem 33 (Candidate stable-phase spectrum audit). *For the candidate*

$$\Phi_{\mathcal{M}} = \sqrt{n} e^{i\theta} \chi,$$

write

$$n = n_0 + \eta, \quad \theta = \nu_0 \tau + \vartheta, \quad \chi = \chi_0 + \xi.$$

The quadratic spectrum should split schematically as

$$\delta^2 S_{\mathcal{M}} = S_{\eta\eta}^{(2)} + S_{\vartheta\vartheta}^{(2)} + S_{\xi\xi}^{(2)} + S_{\text{mix}}^{(2)}.$$

The desired Root A pattern is:

sector	required low-energy status
η	gapped, constrained, near-zone or suppressed
ϑ	common rhythm/cone variable, not by itself the graviton
ξ	internal/confined/gauge/gapped/suppressed or shared-cone shear/matter
S_{mix}	renormalization only; no extra cone

The phase branch may have

$$\omega^2 = c_{\text{rec}}^2 k^2 + O(k^4), \quad G^{\mu\nu} k_\mu k_\nu = 0,$$

and can serve as the candidate common record cone. But this is only Root A input. It does not by itself prove gravitational dynamics:

$$\text{one cone} \neq \text{two tensor gravitons}.$$

If RMG identifies the phase/acoustic branch itself with the graviton, it risks collapsing into a scalar acoustic-gravity model. The correct local-GR route is

$$\text{one common record cone} \xrightarrow{\text{Root B}} \text{ADM constraints} \xrightarrow{\text{two-derivative fixed point}} S_{\text{EH}} \Rightarrow \text{two tensor gravitational modes}.$$

Therefore this candidate passes the first spectrum audit only if the amplitude mode is not a long-range scalar gravity mode, internal modes do not leak extra cones, the acoustic cone is universal for record matter, and the tensor sector is derived by Root B rather than assumed.

Open Theorem 34 (Composite record metric and spin-2 emergence gate). *The record metric must be a composite response of the base medium:*

$$g_{\text{rec}}^{\mu\nu} = \mathcal{G}^{\mu\nu}[\Phi_{\mathcal{M}}], \quad g_{\text{rec}}^{\mu\nu} = \bar{g}^{\mu\nu} + \delta g_{\text{rec}}^{\mu\nu}.$$

The phase/acoustic branch may define the common cone, but the gravitational field is the low-energy dynamics of the composite record geometry, not simply the scalar phase ϑ :

$$\delta g_{\text{rec}}^{\mu\nu} = \frac{\delta \mathcal{G}^{\mu\nu}}{\delta \Phi_{\mathcal{M}}} \delta \Phi_{\mathcal{M}} + \dots$$

In spatial language,

$$\delta h_{ij} = h_{ij}^{TT} + \partial_i V_j + \partial_j V_i + \left(\partial_i \partial_j - \frac{1}{3} \delta_{ij} \nabla^2 \right) B + \frac{1}{3} \delta_{ij} H.$$

Root B must make the scalar and vector pieces constraint, gauge or near-zone, leaving only

$$\boxed{h_{ij}^{TT}}$$

as propagating vacuum gravitational data:

$$\boxed{\#(\text{propagating vacuum gravitational DOF}) = 2.}$$

Possible microscopic sources of this tensor/shear sector are anisotropic stress/strain of χ , collective deformation of the record propagation kernel $G^{\mu\nu}$, topological/oscillon-network elasticity, or emergent embedding geometry after quotienting record histories. None may create a second long-range matter cone or private spacetime. Thus:

$$\boxed{\text{acoustic cone} = \text{kinematics}, \quad \text{ADM closure} = \text{gravitational dynamics}.}$$

If the candidate supplies only a scalar acoustic mode, it fails the spin-2 gate.

Gate 4 (Tensor-closure reduction). The spin-2 issue is not closed in RMG-1. It is reduced to a finite calculation on the candidate record action. The candidate may pass only if the record-visible χ -strain has a transverse-traceless projection and its quadratic kinetic matrix has exactly two positive, luminal, common-cone radiative modes. Equivalently:

$$\boxed{\text{candidate tensor closure} \iff \text{rank } K_{\chi\text{-strain}} = 2 \text{ on the physical record sector}.}$$

Rank zero means no tensor gravity; rank greater than two means extra local shear/scalar/vector radiation; a wrong sign, mass term, or different cone also fails. The detailed proof attempt and projector algebra belong in the work dossier, not in the core file, until the kinetic-rank test is actually derived from $S_{\mathcal{M}}$ or $S_{\text{eff}}^{\text{rec}}$.

This is the central theorem of RMG. It is not yet proven in RMG-1.

14 Allowed Deviation Windows

RMG may develop extra sectors only if they do not break the local GR core.

Allowed windows:

MOND-like low-acceleration response,
cosmological background state of the medium,
matter-sector emergence,
high-energy corrections.

Rule:

$$\boxed{\text{deviations must not spoil the local universal metric, } \gamma = \beta = 1, \text{ or } c_{\text{GW}} = c.}$$

They also must not introduce an independent local $u_{\mathcal{M}}^{\mu}$ into the tested core.

Gate 5 (Flow/vorticity split). *Medium flow and vorticity are allowed only in two protected roles. In the tested local regime they must be hidden inside ADM constraint variables or near-zone fields:*

$$\boxed{A_i, u \quad \text{constraint/near-zone only, not independent vacuum radiation.}}$$

In large-scale or low-acceleration regimes they may re-enter as an effective medium correction:

$$\boxed{\mathbf{a} = \mathbf{a}_{\text{GR}} + \mathbf{a}_{\text{med}}.}$$

The correction must satisfy:

$$\boxed{\mathbf{a}_{\text{med}} \rightarrow 0 \quad \text{in local tested high-acceleration regimes,}}$$

while it may become nonzero in galactic low-acceleration or cosmological background regimes.

Gate 6 (Local flow decoupling). *In the tested local core, medium flow and vorticity may enter only as constraint, shift, near-zone or lower-order transport data. They must not modify the physical record-sector principal tensor:*

$$\boxed{\Delta P_A^{\text{local}}(k) \not\propto \alpha_A v^\mu v^\nu k_\mu k_\nu \quad \text{for external record-forming } A.}$$

Equivalently, the preferred-frame PPN target is:

$$\boxed{\alpha_1 = \alpha_2 = \alpha_3 = 0 \quad \text{at the GR/local fixed point.}}$$

Large-scale medium corrections are allowed only through a reentry function $\mathcal{F}$ satisfying:

$$\boxed{\mathcal{F} \rightarrow 0 \quad \text{in lab, Solar-System, binary-pulsar and local GW regimes.}}$$

This gate is open from $\mathcal{M}$; if it fails, flow/vorticity becomes an observable preferred frame.

Open Theorem 35 (MOND-safe reentry). *Derive a scale or acceleration gate:*

$$\boxed{\mathbf{a}_{\text{med}} = \mathcal{F}\left(\frac{|\nabla\Phi|}{a_0}, \frac{L}{L_{\text{bg}}}, \omega_{\text{flow}}\right) \mathbf{a}_{\text{flow}},}$$

with:

$$\boxed{\mathcal{F} \rightarrow 0 \quad \text{in local tested regimes,} \quad \mathcal{F} \neq 0 \quad \text{in galactic low-acceleration regimes.}}$$

The exact function $\mathcal{F}$ is not fixed in RMG-1.

15 Strong-Field Compact-Object Contract

RMG-1 does not yet select the strong-field compact-object branch. The branch must be derived from the base-medium equilibrium law and the record/internal projection, not imported from the old ISPG bi-conformal scalar construction.

The currently allowed branches are:

$$\boxed{\text{GR/Kerr-like exterior,} \quad \text{old-like exponential branch,} \quad \text{hybrid branch,} \quad \text{or a new branch.}}$$

Old ISPG strong-field results, such as horizonless static exteriors, exponential freezing, curvature-soft deep boundaries, shifted ISCOs and breathing-mode tests, are retained only as candidate targets. They are not RMG-1 predictions unless they are rederived from $\mathcal{M}$ without a fundamental phantom scalar.

Gate 7 (Strong-field compact projection). *For compact objects, the internal sector may dominate bound medium structure and contribute to total energy:*

$$E_{\text{total}} = E_{\mathcal{R}} + E_{\mathcal{I}} + E_{\text{bind}}.$$

However, the external observable cone must be defined by the record projection:

$$\Pi_{\mathcal{R}} P_{\text{full}}(k) \Pi_{\mathcal{R}} = (G^{\mu\nu} k_{\mu} k_{\nu}) K_{\mathcal{R}},$$

while internal compact modes may enter the gravitational source only through conserved total charges such as energy, angular momentum and allowed gauge charges. They must not define an independent external spacetime cone.

Open Theorem 36 (Hybrid compact-object program). *Derive, or rule out, the hybrid strong-field branch:*

$$\text{external record geometry is GR/Kerr-like to current precision}$$

while

$$\text{deep internal base-medium structure may freeze, soften or store energy.}$$

The required proof is:

$$\Pi_{\mathcal{I}} \text{ contributes to } H_{\text{phys}} \quad \text{but} \quad \Pi_{\mathcal{I}} \text{ does not create a separate external cone.}$$

If this fails, RMG must either reduce to a fully GR-like strong-field branch or derive and observationally confront an old-like/new strong-field branch.

16 Statement Status Map

Statement	Status
one base-medium $\mathcal{M}$	Primitive postulate
stable low-energy phase	Primitive postulate
minimal operational node scale ℓ_*	Primitive postulate / Planck-order calibration
Operational Relational Covariance	Primitive protection principle
positive physical energy	Primitive postulate
local effective hierarchy	Primitive EFT condition
no independent $u_{\mathcal{M}}^\mu$	Protected core requirement
diverse local tail imprints in one base-medium	Allowed internal/species structure
$\mathcal{R}$ record sector	Definition / projection still open from $\mathcal{M}$
$\mathcal{I}$ internal sector	Definition; contributes energy, not external core
Root A: one external record carrier channel	Root open theorem
Root A spectral/resonance closure	Open calculation; one gapless branch
Root B: record-embedding ADM closure	Root open theorem
Root B embedding/symplectic closure	Open calculation; null directions $\Rightarrow$ HDA
Root B finite sufficiency checklist	Reduced to B1–B5 plus $A[N, M] = 0$
Root B intuition audit	B1 strong; B2 depends on Root A; B3–B5 open
record phase space $\Gamma_{\mathcal{R}}$ and $\Omega_{\mathcal{R}}$	Construction program; Noether moment-map route
background-free record-action gate	Noether split pass/fail test
two-derivative record-action fixed point	EH if one metric and no background leak
Root C: total-energy source projection	Root open theorem
Root C finite source chain	Reduced to C1–C5; source anomaly gate
C3 lapse-variation source theorem	Conditional proof of $S_u = \rho_E$
no explicit species-private u -coupling	Metric-exhaustion / fifth-force gate
non-circular proof order	Root A $\Rightarrow$ Root B $\Rightarrow$ EH $\Rightarrow$ Root C
$S_{\text{eff}}^{\text{rec}}$ derivation pipeline	Base-medium $\Rightarrow$ record action target
relational base action candidate	Leading-order core route under explicit condition
loop-defect curvature route	Leading-order $\Omega_{\text{loop}} \Rightarrow R(h)$
DeWitt kinetic route	Leading-order shear/constraint split
local ADM/EH core from $\mathcal{M}_{\text{rel}}$	Conditional leading-order result
minimal base-medium input package $\mathcal{D}_{\mathcal{M}}$	Required before microscopic proof
candidate base-medium order parameter $\sqrt{n}e^{i\theta}\chi$	Test ansatz; needs record completion
coherent endpoint-counting square-law route	Completion theorem for stable record sector
stable-record coefficients $C_t : C_0 : C_x = 1 : \rho : \rho^2$	Explicit completion result
candidate stable-phase spectrum audit	Phase cone candidate; tensor gravity still Root A
composite record metric/spin-2 gate	Requires δg_{rec} with protected TT sector
tensor-closure reduction	Open finite test: $\text{rank } K_{\chi-\text{strain}} = 2$
local gravity core closure package	Conditional on Roots A–C
constraint protection route	Open theorem; scalar/vector no-radiation route
two tensor gravitational DOF	Target if first-class constraints close
GW common-mode self-calibration	Protection principle / Michelson–Morley analog
GW detector-signal gate	Open theorem; observable signal must be TT/shear
hypersurface-deformation closure filter	Open ADM algebra gate
DeWitt $\lambda = 1/2$	Selected by closure filter; open from $\mathcal{M}$
EH fixed point	Conditional if ADM closure closes
matter-coupled closure	Open theorem; equivalence-principle gate
species-private source $\sigma_{\text{species}} = 0$	Required source universality
internal energy effacement	Required; energy gravitates without extra core
flow/vorticity local role	Constraint/near-zone only in tested core
local flow decoupling	Open gate; preferred-frame null target
MOND-safe reentry	Open large-scale correction gate

17 What Is Not Yet Proven

RMG-1 is not a closed theory yet.

Still open:

upgrade the leading-order relational core route into an exact microscopic derivation,

construct the exact $S_{\text{eff}}^{\text{rec}}[\mathcal{M}_0]$ from the stable base-medium phase,

specify the input package $\mathcal{D}_{\mathcal{M}} = (\Phi_{\mathcal{M}}, S_{\mathcal{M}}, \mathcal{M}_0, \ell_*, \Omega_{\mathcal{M}}, \mathcal{G}_{\text{int}}, \mathcal{O}_{\text{rec}})$,

test the candidate order parameter $\Phi_{\mathcal{M}} = \sqrt{n}e^{i\theta}\chi$ against Roots A–C,

compute the stable-phase spectrum of the candidate $S_{\mathcal{M}}$,

prove every long-range stable record family uses the coherent endpoint-counting coefficients $C_t : C_0 : C_x = 1 : \rho :$

prove the phase/acoustic branch gives only the common cone, not a scalar graviton,

derive the composite map $g_{\text{rec}}^{\mu\nu} = \mathcal{G}^{\mu\nu}[\Phi_{\mathcal{M}}]$,

prove δg_{rec} contains a protected h_{ij}^{TT} sector,

close the tensor kinetic-rank test: exactly two positive luminal TT modes and no extra tensor/scalar/vector radi

prove amplitude/internal modes are gapped, constrained, gauge, confined or suppressed,

define projectors $\Pi_{\mathcal{R}}, \Pi_{\mathcal{I}}$ and derive the record/internal split,

prove the stable-phase spectrum has exactly one external gapless record branch,

integrate out I and prove no local background leak remains in $S_{\text{eff}}^{\text{rec}}$,

derive $Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB}$ from the record kinetic kernel,

prove the roots in the non-circular order: $A \Rightarrow B \Rightarrow \text{EH} \Rightarrow C$,

prove Root A: one common external record carrier from the stable phase,

compute $\text{Spec}(\mathcal{L}_{\mathcal{M}_0})$ and prove one gapless record branch,

derive phase locking of stable tails to ν_0 or its harmonic channel,

prove Root B: record-embedding first-class ADM closure,

derive record-history equivalence as a true symplectic gauge redundancy,

prove embedding-composition HDA with the same h^{ij} as record geometry,

derive lapse/shift multiplier status from base-medium embedding redundancy,

verify B1–B5 for Root B in the base-medium record sector,

prove the Root B anomaly gate $A[N, M] = 0$,

exclude unsuppressed lapse/shift scalar-vector kinetic terms in the tested core,

construct the low-energy record phase space $\Gamma_{\mathcal{R}}$,

construct the record presymplectic form $\Omega_{\mathcal{R}}$,

derive embedding vector fields δ_{ξ} and their local generators D, H ,

derive the Noether split $J_{\xi} = C_{\xi} + dQ_{\xi}$ for the record action,

show every bulk part of C_ξ is a constraint, not a physical charge,
 prove the local record action is background-free: $L_{\mathcal{R}}^{\text{local}} = L[g_{\text{rec}}, \Psi_{\text{rec}}]$,
 classify every Noether bulk remainder and prove $\mathcal{A}_\xi^{\text{local}} = 0$,
 derive the two-derivative record-action EFT basis from $\mathcal{M}$,
 prove every local background-leak operator is absent/constrained/gauge/suppressed,
 prove Root C: total-energy source projection and far-tail Gauss law,
 verify C1–C5 for Root C in the base-medium matter sector,
 prove $S_u = \rho_E$ for every stable record matter family,
 prove stable matter has no explicit unsuppressed species-private u -coupling,
 derive metric-exhaustion of local u -dependence from stable matter construction,
 show any $F_A(u, \Psi_A)$ is absent, suppressed, internal, gauge or already in ρ_E ,
 prove internal/binding/confined energy is included in $\mathcal{H}_m$,
 prove $\sigma_{\text{species}}^{\text{long, local}} = 0$,
 derive Operational Relational Covariance from microscopic $\mathcal{M}$,
 derive the record/internal projection $\Pi_{\mathcal{R}}$ from $\mathcal{M}$,
 prove forbidden preferred-frame operators are not generated,
 derive the factorization gate $Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB}$,
 prove the stable record phase has only one observable principal tensor,
 derive non-common mode elimination: confined/gapped/damped/gauge/unstable,
 prove the observable tensor basis gate in the tested local core,
 eliminate unsuppressed $U^\mu U^\nu$ and flow/vorticity tensors locally,
 derive local flow decoupling and preferred-frame PPN null behavior,
 prove no physical low-energy birefringence in the record phase,
 derive the stable-tail common-carrier theorem for record-forming oscillons,
 derive record-sector no-birefringence from one stable record phase,
 derive one-cone clock universality $s_A/s_B = \text{constant}$,
 prove the failure matrix $B_{AB}^{\mu\nu}$ vanishes on physical principal modes,
 derive universal matter coupling from microscopic modes,
 derive matter-coupled ADM closure and $\nabla_\mu T^{\mu\nu} = 0$,
 prove no unsuppressed species-private gravitational source remains,
 prove internal/binding energy effaces into total energy and multipoles,
 derive the universal gravitational tail projection $q_A \propto E_A/c^2$,
 derive the far-tail monopole/source density $Q_A^{(0)} = \eta E_A/c^2$,

derive the universal operational factor s for the full stable spectrum from $\mathcal{M}$,
 derive the coherent endpoint-counting record completion uniquely from the pre-continuum node dynamics,
 derive ℓ_* from the stable base-medium record dynamics or keep it as primitive calibration,
 prove the residual medium term $\Delta_{\mathcal{M}}$ vanishes or is suppressed,
 derive scalar/vector constraints and tensor/shear radiation,
 derive lapse/shift no-kinetic structure from $\mathcal{M}$,
 prove pure common-mode breathing is locally self-calibrated and detector-null,
 derive the detector-visible TT/shear or defect projection for GW signals,
 prove first-class constraint closure removes scalar/vector radiation,
 derive the hypersurface-deformation algebra from record embeddings,
 derive the DeWitt coefficient $\lambda = 1/2$ from the protected phase,
 prove $\Delta_{\mathcal{M}} = 0$ or constrained/gauge/suppressed in the closure filter,
 derive ADM closure and Einstein-Hilbert fixed point,
 control preferred-frame effects,
 derive the flow/vorticity gate and MOND-safe reentry function $\mathcal{F}$,
 derive the compact-object branch and strong-field projection gate,
 rebuild MOND, cosmology and matter sectors without a phantom foundation.

18 Pass/Fail Gates

Gate 8 (Core pass). *RMG passes as a core framework if it derives:*

$$S_{\text{EH}} + S_{\text{matter}}$$

as the low-energy fixed point of $\mathcal{M}$, with positive physical Hamiltonian, universal metric coupling, and only two vacuum tensor gravitational degrees of freedom.

Gate 9 (Core fail). *RMG fails if the base-medium produces an observable local preferred frame, species-dependent metric coupling, extra scalar/vector radiation, wrong gravitational-wave speed, negative-energy physical modes, or non-closing ADM constraints.*

Gate 10 (GW detection fail). *RMG fails in the tested local gravitational-wave sector if observed interferometer signals are explained only by a pure common-mode scalar breathing that should be self-calibrated away, if an unsuppressed breathing/scalar polarization survives local polarization tests, or if no protected differential TT/shear record signal is produced.*

Gate 11 (Tensor-closure fail). *RMG fails in the tested local tensor sector if the candidate record dynamics does not produce exactly two positive, luminal, common-cone TT modes, or if it leaves an unsuppressed scalar/vector, massive spin-2, negative-energy, or extra shear radiation channel.*

Gate 12 (Preferred-frame fail). *RMG fails in the tested local core if an independent $u_{\mathcal{M}}^{\mu}$ enters local operators, or if lapse/shift acquire low-energy kinetic terms.*

Gate 13 (Factorization fail). *RMG fails in the tested local core if physical low-energy principal operators do not factorize through one $G^{\mu\nu}$, or if the failure matrix $B_{AB}^{\mu\nu}$ produces species-dependent cones, a preferred-frame tensor, or a geometry cone different from the matter cone.*

Gate 14 (Tail-projection fail). *RMG fails in the tested local core if stable low-energy particles have different gravitational projection ratios $q_A/(E_A/c^2)$, or if their long tails require different long-range spacetime carriers.*

Gate 15 (Medium-residue fail). *RMG fails in the tested local core if $\Delta_{\mathcal{M}}$ leaves an unsuppressed preferred-frame term, extra operational tensor, non-ADM constraint bracket, or low-energy scalar/vector propagating degree of freedom.*

Gate 16 (MOND-window fail). *RMG fails if the MOND-like medium correction remains unsuppressed in solar-system, binary-pulsar or gravitational-wave local tests, or if it changes $\gamma = 1$, $\beta = 1$, $c_{\text{GW}} = c$, universal free fall, or the two-tensor-mode vacuum sector.*

Gate 17 (Strong-field compact fail). *RMG fails in the compact-object sector if internal medium modes hide energy from gravity, create an unsuppressed breathing/scalar radiation channel, define a species-private external cone, alter $c_{\text{GW}} = c$, or force shadow, ringdown, binary-pulsar or strong-field lensing deviations already excluded by observations.*

Current Status

RMG-1 has a conditional leading-order local GR core, not yet a closed theory.

Main theorem to prove:

$$\mathcal{M} \Rightarrow \text{exact ORC/no-extra-tensor} \Rightarrow Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB} \Rightarrow \Delta_{\mathcal{M}} = 0 \Rightarrow \text{ADM/Einstein-Hilbert low-energy fixed poi}$$